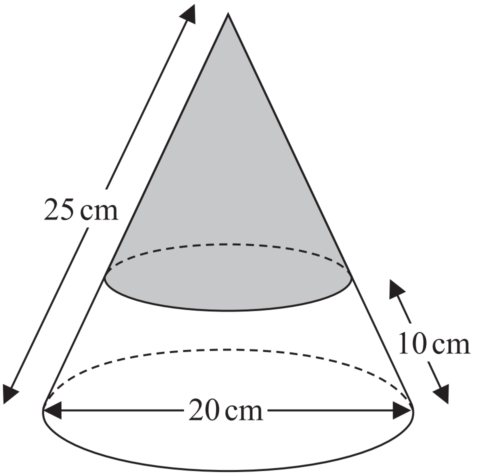
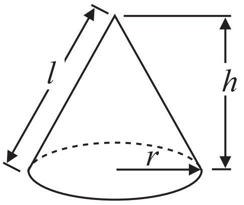
18 The diagram represents a solid cone.



Curved surface area of cone = πrl



The cone has a base diameter of 20 cm and a slant height of 25 cm.

A circle is drawn around the surface of the cone at a slant height of 10 cm above the base.
The curved surface of the cone above the circle is painted grey.

Work out the area of the curved surface of the cone that is **not** painted grey.
Give your answer as a multiple of π
You must show all your working.

Total curved area = $\pi \times 10 \times 25 = 250\pi$ cm^2.

Grey top cone has slant height $25 - 10 = 15$ cm.

Similar cones: small radius = $10 \times 15/25 = 6$ cm.

Grey curved area = $\pi \times 6 \times 15 = 90\pi$ cm^2.

Unpainted area = $250\pi - 90\pi = 160\pi$ cm^2.

4 marks total
Whole cone 250π
Small radius 6
Subtract small area 90π
160π cm^2

.... cm^2

(Total for Question 18 is 4 marks)

DO NOT WRITE IN THIS AREA
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